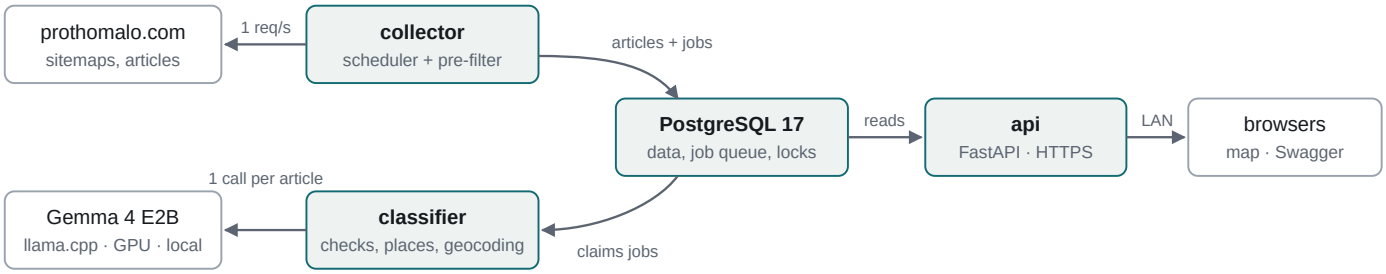
# Dhaka Crime Map — Pipeline Report

26 October 2026 · in-house deployment on **genweb2inhouse** (192.168.1.200) · repository github.com/zawadgenweb2/dhaka-crime-map · status: **live**

## 1. What it does

- Collects crime news from **Prothom Alo** (Bangla) twice a day (08:00 and 20:00 Asia/Dhaka), plus on-demand backfills of past days.
- An **in-house language model (Gemma 4 E2B)** reads each likely crime article and extracts incidents: crime type, place in Dhaka district, date, time of day, victims and a verbatim supporting quote.
- Incidents that pass automatic checks appear on a bilingual (Bangla/English) **live heat map** with filters, hotspots and a news feed; every point links to its source article. No external AI service is used.

## 2. Architecture




Five Docker services (db, migrate, api, collector, classifier) plus one systemd model server, all on one in-house GPU server (RTX 3080 Ti, 12 GB). Postgres is the database, job queue and lock service; the database exists only on this server.

## 3. Pipeline, step by step

| # | STEP | HOW | RESULT |
|---|---|---|---|
| 1 | Discover | Prothom Alo news sitemap (last 48 h, with titles) + one daily sitemap per day; robots.txt honoured; 1 request/s | ~300 articles/day listed |
| 2 | Pre-filter, before download | Crime & capital sections always; any other section if the **title or keywords contain a crime word**; videos, galleries, sister sites never | ~10% downloaded |
| 3 | Pre-filter, full text | Needs a crime word and a **Dhaka-district place** in the text (namesakes like "কুষ্টিয়ার মিরপুর" and "sent to Dhaka Medical" do not count) | ~20–30 articles/day to Gemma |
| 4 | Classify | One Gemma call per article: full text + the Dhaka places it mentions + an 8-day Bangla calendar; answer forced into a JSON schema | incidents, ~3 s/article |
| 5 | Check & locate | Quote must be verbatim in the article; place must be one the article names; first sentence, quote sentence and location phrase must not put it in another district; no future dates | published or held |
| 6 | Group duplicates | Same category + place, dates ±1 day, compatible time of day | one point per event |
| 7 | Exact spot | Street/landmark geocoded via OpenStreetMap (1 request/15 s), kept only within 2.5 km of the place | refined points |
| 8 | Serve | REST API + dashboard; the page updates itself when data changes; per-day coverage shows how complete a period is | live map |

## 4. Model and quality

**Gemma 4 E2B** (Q8 GGUF) on **llama.cpp**, CUDA, loopback only with an API key, always loaded (2.9 GB GPU memory). Measured on 71 reference articles labelled in the prototype phase:

| MEASURE | RESULT |
|---|---|
| Valid structured answers | 100% |
| Article: crime in Dhaka? precision / recall | 0.87–0.90 / 0.81 |
| Incidents: precision / recall | 0.80–0.83 / 0.64–0.66 |
| Thana / place / category / date correct | 0.93 / 0.86 / 0.76 / 0.55–0.63 |
| Time per article (median) | 2.4 s |

**Engine/model benchmark** (same articles and prompt):

| SETUP | INCIDENT P / R | DECODE | GPU MEM |
|---|---|---|---|
| **E2B, llama.cpp** (chosen) | 0.80–0.83 / 0.65 | 120 tok/s | 2.9 GB |
| E4B (QAT Q4), llama.cpp | 0.71 / 0.69 | 113 tok/s | 3.5 GB |
| E4B (QAT w4a16), vLLM | stopped | ~120 tok/s | 11.7 of 12 |

vLLM gave no speed gain for one-at-a-time requests and needed nearly the whole card; the larger E4B invented more incidents. Prefill ≈ 4,400 tok/s; first token ≈ 0.14 s.

## 5. Current status (26 Oct, 16:00 Dhaka)

| 285 | 198 | 620 | 24 MB | 76 |
|---|---|---|---|---|
| incidents on the map | in the last 30 days | articles read by Gemma | database size | automated tests passing |

The **30-day backfill is complete** (27 Sep – 26 Oct): 2,800 articles checked, 620 read by Gemma, 0 failures. Coverage: 28 days complete; 2 days each miss one page that has no article text (a style/fashion page). All incidents were re-checked with the final location rules from Gemma's saved answers (no new model calls); 47 extracted incidents are held back (quote not in the article, no Dhaka place, or another district). Prototype (Claude) output was removed from the map and is kept only as a frozen test set. Page 3 summarises what the data shows.

## 6. Safeguards

- **Only in-house Gemma output is served**; every endpoint checks provenance.
- **Evidence:** each incident carries a quote that must appear word for word in the article.
- **Location grounding:** the model may only pick Dhaka places the article names; another district in the first sentence, the quote's sentence or the location phrase blocks publishing ("sent to Dhaka Medical" does not count as Dhaka).
- **Reliability:** every model call is recorded before it is sent and every answer saved, so re-checking needs no new model calls; an older job can never overwrite newer output.
- **Coverage honesty:** each day is marked complete, processing, partial, failed or not collected; the map warns when a period is incomplete.
- **Politeness:** robots.txt, 1 request/s to the publisher, OpenStreetMap ≤ 4/min.
- **Security:** LAN-only HTTPS; admin token for actions that change data or use the GPU; secrets never in git.
- **Backups:** nightly database dump, 14 days kept.

## 7. Interfaces

| READ (OPEN ON THE LAN) | ADMIN (TOKEN) |
|---|---|
| `/api/incidents`, `/api/stats` (filters: days or dates, crime type, thana) · `/api/coverage` per-day completeness · `/api/meta` status and progress · `/api/runs` history | `POST /api/scrape` collect now (up to 90 days) · `POST /api/classify` try any pasted article, nothing saved · review: fix category/place, reject, retry, dismiss an article |

Dashboard https://192.168.1.200:8100 · Swagger https://192.168.1.200:8100/docs · API guide with real responses: docs/API.md · engineering handbook: docs/ENGINEERING.md.

## 8. Known limits

- One newspaper; these are **reported** incidents, not all crime. Results are published without human review, so some errors reach the map (incident precision ≈ 0.8).
- The model still sometimes splits one event into several incidents (e.g. extortion + kidnapping + assault for one case); dates are the weakest field.
- Older days: Prothom Alo's daily lists have no titles, so past sports/business/world articles are not examined (a 48-hour audit found no Dhaka crime among them).
- Points without a named street sit at the neighbourhood or thana centre. Browser shows "Not secure" (self-signed LAN certificate) until Tailscale HTTPS is enabled.

# Dhaka crime, last 30 days — what the pipeline found

27 September – 26 October 2026 · Prothom Alo, Dhaka district · 198 incidents after duplicate grouping · extracted by Gemma, not human-verified

| 198 | ~6.6 | 39% | Shah Ali | 28 / 30 |
|---|---|---|---|---|
| reported incidents | per day on average | at night (21:00–05:00) | most incidents (14) | days fully covered |

## Crime types

| | | |
|---|---|---|
| Assault & clashes | 43 | |
| Murder | 39 | |
| Drugs & weapons | 29 | |
| Theft & burglary | 29 | |
| Fraud & extortion | 26 | |
| Kidnapping | 17 | |
| Other | 7 | |
| Sexual violence | 6 | |
| Mugging & robbery | 2 | |

## Time of day

| WHEN | INCIDENTS | SHARE |
|---|---|---|
| Night (21:00–05:00) | 78 | 39% |
| Afternoon (12:00–17:00) | 54 | 27% |
| Morning (05:00–12:00) | 25 | 13% |
| Evening (17:00–21:00) | 18 | 9% |
| Not stated | 23 | 12% |

## Hotspots (thana)

| THANA | INCIDENTS | MAIN TYPES |
|---|---|---|
| Shah Ali | 14 | assault 7, theft 4 — mostly the Raddha health-centre attack |
| Mirpur | 12 | assault 3, theft 3, extortion 2, murder 2 |
| Mohammadpur | 12 | kidnapping 5, theft 4 |
| Cantonment | 10 | theft 5, assault 4 — mostly the New Guinea Properties attack |
| Gulshan | 10 | drugs/weapons 3, mixed |
| Adabor | 10 | extortion 3, mixed |
| Darus Salam | 8 | extortion 3, murder 2 |
| Vatara | 6 | drugs/weapons 4, murder 2 |

## Incidents per day




Weekly: 62, 53, 41, 35, then 7 in the final partial week. Today (26 Oct) has no events yet: articles about it appear later.

## Notable events

- **Raddha MCH-FP health centre, Mirpur/Shah Ali (late Sept):** a crowd demolished and looted a mother-and-child health centre; reported in many follow-up articles, so it appears several times under assault, theft and other.
- **New Guinea Properties / ECB Chattar, Cantonment (26 Sept):** armed attack and vandalism at a housing project and a content creator's studio, with arrests and bail hearings through October.
- **Killings in October:** a Youth Women League leader in Keraniganj, a housewife in Nakhalpara (husband arrested), a security guard in Bashundhara, a businessman in Vatara, a body under Rampura Bridge, a father stabbed by his son in Demra.
- **Garment workers and police clash, Uttara (17 Oct)**, dozens injured; **drug raids** in Gendaria and Vatara.

## How to read these numbers

- **Reported, not recorded crime:** one newspaper's coverage; big stories are reported more often than ordinary crime.
- **Counts are inflated by follow-ups:** court hearings, arrests and bail stories about one event can each add an incident when the model assigns a different type or place (Raddha, New Guinea). A manual check of 12 "murder" items found about 7 new killings in the period; the rest were court orders, threats or older cases.
- **The falling trend is mostly an artefact:** recent events have had fewer follow-up stories so far, and events are dated by when they happened.
- **Victim numbers are unreliable** (the model sometimes counts the attacking crowd) and are not reported here.
- **Location precision:** 128 incidents at thana level, 39 at neighbourhood level, 31 at an exact spot.